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Countering Intelligence Algorithms

Decision Theory, Design Choices and Counter-AI

Peter J Phillips and Gabriela Pohl

The use of artificial intelligence (AI) and algorithms by intelligence agencies requires the capability to counter their use by rival agencies. A formal-technical development of this capability can be assisted by decision theory, including behavioural economics. Peter J Phillips and Gabriela Pohl argue that decision theory provides a useful framework that can be used to think about thinking. It is a catalogue of things that human decision-makers can consult when considering their own decision-making processes or those of adversaries and friends. Such a catalogue may be relevant during the development of AI and counter-AI capability in an intelligence setting.

Information washes over and through contemporary society. Intelligence agencies use technology, including algorithms and artificial intelligence (AI),[1] to collect it, analyse it and even contribute to its flow.[2] They also use technology to counter the use of algorithms and AI by rival agencies. These activities might be seen as purely technical matters but the authors of this article believe that decision theory, especially behavioural economics, can be used by intelligence agencies as a framework with which to think about the human decision-making processes applied during the design and use of their own technologies as well as the human decision-making processes that might have shaped the design and use of algorithms and AI by rival agencies. This article contends that designing effective algorithms and AI, and effectively countering those of rival agencies are tasks that can be guided and aided by some of the basic concepts and principles of decision theory. In this, there remains a role for the human intelligence officer and traditional intelligence in searching for vulnerabilities in rival agencies' algorithms and AI.

Algorithms and AI can be used for information gathering and analysis on the one hand and the malicious production of disinformation – reminiscent of the old Soviet-style active measures (*aktivnyye meropriyatiya*) – on the

1. There is an intimate connection between algorithms and artificial intelligence (AI). According to Kaya Ismail, 'An algorithm is a set of instructions — a preset, rigid, coded recipe that gets executed when it encounters a trigger. AI on the other hand — which is an extremely broad term covering a myriad of AI specialisations and subsets — is a group of algorithms that can modify its algorithms and create new algorithms in response to learned inputs and data as opposed to relying solely on the inputs it was designed to recognise as triggers. This ability to change, adapt and grow based on new data, is described as intelligence ... The key difference, is that an algorithm defines the process through which a decision is made, and AI uses training data to make such a decision'. Kaya Ismail, 'AI Vs. Algorithms: What's the Difference?', CMSWire, 26 October 2018, <https://www.cmswire.com/information-management/ai-vs-algorithms-whats-the-difference/>, accessed 2 June 2020. One way of thinking about AI is as a machine or computer program that has the ability to achieve goals – such as taking a collection of documents and categorising them by topic.
2. See Aaron F Brantly, 'When Everything Becomes Intelligence: Machine Learning and the Connected World', *Intelligence and National Security* (Vol. 33, No. 4, 2018), pp. 562–73; James L Regens, 'Augmenting Human Cognition to Enhance Strategic, Operational and Tactical Intelligence', *Intelligence and National Security* (Vol. 34, No. 9, 2019), pp. 673–87.

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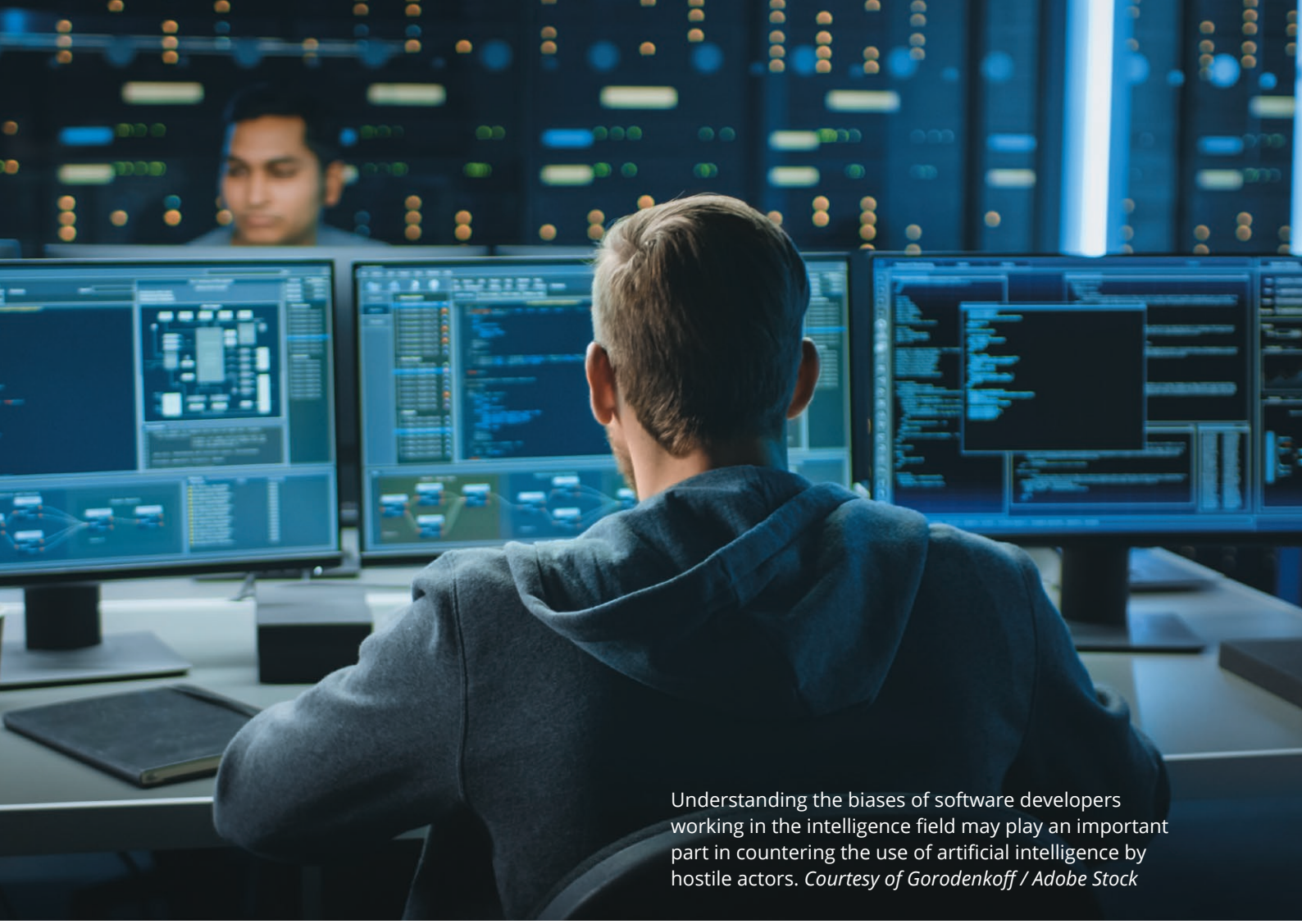

Understanding the biases of software developers working in the intelligence field may play an important part in countering the use of artificial intelligence by hostile actors. *Courtesy of Gorodenkoff / Adobe Stock*

other.[3] But the applications go far beyond analysis and disinformation and much deeper into the information structure of the intelligence setting. For example, consider one of the major challenges of spy-craft: producing cover identities. Edward Lucas explains how cover identities are more easily detected by facial recognition technology (driven by AI) together with algorithms and other technologies that can be used to identify gaps or inconsistencies with online profiles created as part of the cover identity.[4] On the flipside, the task of developing online histories for intelligence officers can be partially outsourced to algorithms and AI, including the production of text, images and video. This can produce realistic material in such volumes that it becomes much more difficult to identify cover profiles by their relative lack of detail and history.

Consider Reuters, which started using algorithms and AI to analyse text, assisting journalists to identify stories. Reuters now uses algorithms to *produce* written news reports. Using algorithms to produce mundane news stories (for example, summaries of companies' financial reports) frees journalists to engage with other journalistic pursuits. Interestingly, Reuters is even experimenting with the use of algorithms and AI to produce *video* news reports for sporting events. The algorithms develop a script based on the flow of information about a game. Eerily, this is then 'read' by a digital copy of a human presenter. The digital presenter is controlled by an AI system that has been programmed to simulate the human newsreader.[5]

When it comes to everyday intelligence analysis, Joseph W Gartin highlights the centrality of 'read stuff, write stuff' tasks.[6] Some commentators see algorithms taking the place of these traditional functions, but the reality is much more nuanced. According to Wilson Andrews and Todd Lindeman, across all relevant agencies and departments in the US,[7] annual data collection expenses total more

3. See Dennis Kux, 'Soviet Active Measures and Disinformation: Overview and Assessment', *Parameters* (Vol. 15, No. 1, 1985), pp. 19–28.
4. Edward Lucas, 'The Spycraft Revolution', *Foreign Policy*, 27 April 2019.
5. Simon Chandler, 'Reuters Uses AI to Prototype First Ever Automated News Reports', *Forbes*, 7 February 2020.
6. Joseph W Gartin, 'The Future of Analysis', *Studies in Intelligence* (Vol. 63, No. 2, 2019), p. 2.
7. The CIA, National Security Agency, National Reconnaissance Office, National Geo-Spatial Intelligence Program, General Defence Intelligence Program, Department of Justice and Office of the Director of National Intelligence.

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than $20 billion while annual data analysis expenses total around $20 billion.[8] This does not include expenditure on management, facilities and support which at the National Security Agency (NSA) alone totals more than $6 billion annually. By performing routine tasks, AI can free up (rather than usurp) intelligence analysts and allow financial resources to be allocated to higher-level analysis. Joe McCloskey and David J Mountain, in their overview of the use of AI by the NSA, point out, 'Another technology benefiting from AI is the smart digital assistant, with which the intelligence community has tremendous potential to revolutionize analysis by helping analysts find the data and analytics they need, fostering collaborations, and making workflow recommendations'.[9] These innovations are essential because agencies are confronting a 'data deluge'.[10]

Because of unfamiliarity with the ideas underlying the use of algorithms and AI, it is easy to lose sight of a fundamental fact: when someone – a human – designs or builds algorithms or AI, they do so for a purpose. Basic concepts from decision theory and behavioural economics can be used to think about this design process and identify the types of things that shape it in both friendly and adversarial intelligence agencies. For many reasons, there may be a gap between vision and completed design.[11] Drawing on decision theory can help to understand this gap, to plug it in one's own agency or exploit it in another's. Decision theory and behavioural economics provide leads to pursue, places to look and human decision-making processes about which to think more carefully.

One of the foundational concepts of decision theory is bounded rationality. This refers to the limits that apply to the performance of any decision-making task. The limits are partly due to the physiological limits of human cognition and partly due to the limited time and resources that people have to spend on a given task.[12] This leads to a second foundational concept called 'satisficing'. Because of bounded rationality, people satisfice rather than optimise. Rather than work through every possible action and choose the best one, they work through a limited number of actions and choose the best one they can find. Bounded rationality, including limits on resources and data availability, and the accompanying necessity to find satisficing solutions,[13] rather than absolutely optimal solutions, leads this article to suggest that 'problem space delineation' is a key feature of algorithm and AI design. The nature of the problem that the algorithm or AI solves is transformed during problem space delineation, shaping the interplay between vision and completed design.

Beyond bounded rationality and satisficing, decision theory and behavioural economics can be used to guide thinking about other aspects of designers' decision-making processes. Human decision-makers often make use of heuristics. Heuristics are rules of thumb or shortcuts that enable fast thinking.[14] They are usually subconscious and the decision-maker is unaware of their operation. Heuristics might shape the design of algorithms and AI in different ways. Behavioural economics, which is based on Daniel Kahneman and Amos Tversky's

8. Wilson Andrews and Todd Lindeman, 'The Black Budget', *Washington Post*, 29 August 2013.
9. Joe McCloskey and David J Mountain, 'Guest Editors' Column', *Next Wave* (Vol. 22, No. 2, 2019), pp. 1–2.
10. *Ibid.* p. 2.
11. Nathan Lineal, 'Game Theoretic Aspects of Computing', in Robert Aumann and Sergiu Hart (eds), *Handbook of Game Theory*, Vol. 2 (Amsterdam: North Holland, 1994), pp. 1339–95; Bernardo A Huberman and Tad Hogg, 'Distributed Computation as an Economic System', *Journal of Economic Perspectives* (Vol. 9, No. 1, Winter 1995), pp. 141–52; Noam Nisan and Amir Ronen, 'Algorithmic Mechanism Design', *Games and Economic Behaviour* (Vol. 31, No. 1–2, 2001), pp. 166–96.
12. John Conlisk, 'Why Bounded Rationality?', *Journal of Economic Literature* (Vol. 34, No. 2, 1996), pp. 669 –700.
13. Bounded rationality and satisficing are concepts developed by Herbert A Simon. See Herbert A Simon, *Administrative Behaviour: A Study of Decision-Making Processes in Administrative Organisation* (New York, NY: Macmillan, 1947); Herbert A Simon, 'A Behavioural Model of Rational Choice', *Quarterly Journal of Economics* (Vol. 69, No. 1, 1955), pp. 99–118; Herbert A Simon, 'Rationality in Psychology and Economics', *Journal of Business* (Vol. 59, No. 4, 1986), S209–S224; Herbert A Simon, 'Satisficing', in John Eatwell, Murray Milgate and Peter Newman (eds), *The New Palgrave: A Dictionary of Economics* (London: Macmillan, 1987), pp. 243–45; Herbert A Simon, 'Invariants of Human Behaviour', *Annual Review of Psychology* (Vol. 41, 1990), pp. 1–19.
14. Amos Tversky and Daniel Kahneman, 'Judgement Under Uncertainty: Heuristics and Biases', *Science* (Vol. 185, No. 4157, 1974), pp. 1124–31; A Tversky and D Kahneman, 'The Framing of Decisions and the Rationality of Choice', *Science* (Vol. 211, No. 4481, 1981), pp. 453–58; Amos Tversky and Daniel Kahneman, 'Advances in Prospect Theory: Cumulative Representation of Uncertainty', *Journal of Risk and Uncertainty* (Vol. 5, No. 4, 1992), pp. 297–323; Daniel Kahneman and Amos Tversky, 'Prospect Theory: An Analysis of Decision Under Risk', *Econometrica* (Vol. 47, No. 2, 1979), pp. 263–91; Daniel Kahneman, *Thinking, Fast and Slow* (New York, NY: Farrar, Straus and Giroux, 2011).

model of the decision-making process, provides some leads to follow up when thinking about algorithm design within either a friendly or a rival intelligence agency.[15]

In pursuing these themes, this article illustrates, with the aid of a simple example, how a problem space must be delineated in order for an algorithm to work before moving on to explore the ways in which human decision-making processes might shape the design of algorithms and AI. The inner workings of algorithms and AI seem so unfamiliar to the human decision-makers whose decisions they might aid or replace that it would seem strange to suggest that the human roots of algorithms run very deep. But they do. And effectively countering rival algorithms can be guided and aided, therefore, by some of the basic concepts and principles of decision theory.

Problem Space Delineation and Algorithm Design Choices

The CIA's library contains many publicly available examples of human content/text analysis. For example, Document CIA-RDP91-00901R000600200001-2 contains a collection of newspaper articles from the mid-1980s annotated by hand.[16] Even at the time, the flow of information, including news stories, was intense. Now, of course, the flow of information is immense. Martin Hilbert and Priscila López, for example, estimated that in 2007 humankind was able to communicate 2×10^{21} bytes (2 followed by 21 zeroes) or around 3 billion terabytes of information.[17] By 2018, the amount of data created *every day* was estimated at 2.5 quintillion bytes (2.5 followed by 18 zeroes)[18] and it is estimated that this daily creation of data will reach 463 quintillion bytes by 2025.[19] Intelligence is not the only setting in which vast quantities of information must be analysed.[20] The use of algorithms and AI to read, interpret, summarise, categorise and understand this information is now a rapidly expanding enterprise.[21]

With regard to written and spoken human language, natural language processing (NLP) is a multifaceted field of study that encompasses various applications of algorithms and AI to reading and understanding human language. In 2019, the National Security Agency's *The Next Wave* periodical devoted an issue to machine learning, outlining some of the particular ways in which NLP and other technologies can be applied in an intelligence setting.[22] The Latent Dirichlet Allocation (LDA) algorithm, developed by David M Blei, Andrew Y Ng and Michael I Jordan,[23] is a prominent example of an NLP application. This article uses it to explain why decision theory and behavioural economics helps us to think about the human decisions behind the design process.

Consider the task of identifying the topics or themes that a collection of documents has in common and categorising those documents according to topics. A researcher can do this by

15. Kahneman and Tversky, 'Prospect Theory'.
16. CIA, CIA-RDP91-00901R000600200001-2, annotated collection of documents, 1 January 1985, released 15 November 2005, <https://www.cia.gov/readingroom/document/cia-rdp91-00901r000600200001-2>, accessed 1 June 2020.
17. Martin Hilbert and Priscila López, 'The World's Technological Capacity to Store, Communicate, and Compute Information', *Science* (Vol. 332, No. 6025, 2011), pp. 60–65.
18. Bernard Marr, 'How Much Data Do We Create Every Day? The Mind-Blowing Stats Everyone Should Read', *Forbes*, 5 May 2018, <https://www.forbes.com/sites/bernardmarr/2018/05/21/how-much-data-do-we-create-every-day-the-mind-blowing-stats-everyone-should-read/?sh=5c38569e60ba>, accessed 27 January 2021.
19. Jeff Desjardins, 'How Much Data Is Generated Each Day?', World Economic Forum, 17 April 2019, <https://www.weforum.org/agenda/2019/04/how-much-data-is-generated-each-day-cf4bddf29f/>, accessed 27 January 2021.
20. For example, companies are interested in online reviews or 'mentions' about their products. See Shiliang Sun, Chen Luo and Junyu Chen, 'A Review of Natural Language Processing Techniques for Opinion Mining Systems', *Information Fusion* (Vol. 36, 2017), pp. 10–25; Linda W Lee et al., 'Making Sense of Text: Artificial Intelligence Enabled Content Analysis', *European Journal of Marketing* (Vol. 54, No. 3, 2020), pp. 615–44. Patent lawyers are interested in the 'definiteness' requirements for patent claims, determining which requires the analysis and assessment of significant quantities of textual information. See Dean Alderucci and Kevin Ashley, 'Using AI to Analyse Patent Claim Indefiniteness', *IP Theory* (Vol. 9, No. 1, 2020). The examples are endless.
21. Julia Hirschberg and Christopher D Manning, 'Advances in Natural Language Processing', *Science* (Vol. 349, No. 6245, 2015), pp. 261–66.
22. McCloskey and Mountain, 'Guest Editors' Column'.
23. David M Blei, Andrew Y Ng and Michael I Jordan, 'Latent Dirichlet Allocation', *Journal of Machine Learning* (Vol. 3, 2003), pp. 993–1022.

hand: searching a database and following leads, identifying new key words and so forth until all of the relevant articles are collected and organised. Likewise, a team of intelligence analysts can trawl through caches of documents or other material with the same objective in mind. It takes a lot of time to do this, even if the documents are relatively structured to begin with and even if it is known what the documents are mostly about.

It may be the case, of course, that there is little knowledge of what the documents are about. The topics may be 'latent'. The LDA algorithm can be used to identify these latent topics and sort the documents accordingly. If successful, such an application promises great efficiencies. However, as discussed below, the LDA solves a different but analogous problem to the one that a human solves when he or she sorts documents by topic or theme. The problem space must be delineated in certain ways, each an outcome of a decision bounded by human cognition and limited by time, resources and data availability.

The applications of the LDA include use as a way to: (1) categorise large collections of articles or documents; (2) identify important themes in documents from given time periods (that is, what people were talking about at the time); and (3) identify preferences based on past choices – that is, patterns in choices or preferences revealed in a person's or institution's selection of material.[24] If it works well, such an algorithm can save a lot of time and can help identify latent topics that might not be identified by human analysts. But what can decision theory reveal about the design process? Apart from the technical aspects, what should be considered when looking into the design processes of both friends and adversaries? If the design of the LDA algorithm is examined from a decision theory and behavioural economics perspective, the centrality of problem space delineation becomes apparent, with implications for how a task must be transformed in order for the algorithm to work.[25] Problem space delineation is a fingerprint of human design based on human choice. When humans design and build algorithms to a solve problem, they must delineate the problem space to enable a computer to follow the necessary steps towards a solution. In doing so, the designer makes choices. These choices leave an indelible mark on the nature of the algorithm.

Take the example of a very large number of documents. Think for a moment about what a document *is*, how it was written and what a topic or theme *is*, and then about how a human decision-maker would approach the task of finding the topics or themes in the documents and categorising them accordingly. To program a computer to perform an analogous, though not identical task, the documents have to be deconstructed into component parts. These parts are transformed into mathematical objects with which the computer can work.

Blei, Ng and Jordan present a formal discussion, but the general objective of the designers is to take the problem as it stands and transform it in such a way that it can be solved by a computer.[26] Rather than attach any human significance to documents, the algorithm designers proceed by recognising that to a computer a document is simply a 'bag of words'. This is the primary metaphor underlying the problem space delineation that characterises the LDA.

From a deconstructed view of a document as a bag of words, the designers want the computer to identify topics and then assign each document to categories based on those topics. To do this, words, topics and documents must be defined in such a way that a computer can be programmed to work with them. First, words are represented by lists of numbers (vectors). The word 'cat' might be represented by (00100) while 'dog' might be represented by (01000) and 'rain' might be represented by (00001000). A computer can work with these mathematical objects.

Now consider topics. A topic could be 'weather', 'pets', 'universities' or 'sports'. These have particular significance and meaning to human analysts but the LDA designers have to find a way to depict topics in such a way that the computer can work with them. Their choice is to depict topics as combinations of words. For example, the topic 'pets' is a combination of words such as 'cat', 'dog' and 'rain'. Since some

24. The focus in this article is on text but Latent Dirichlet Allocation (LDA) can be used to categorise documents by the images contained in them. Interestingly, however, images are treated as documents and pieces of the image are treated as words. See L Fei-Fei and P Perona, 'A Bayesian Hierarchical Model for Learning Natural Scene Categories', in *Proceedings of the 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, Vol. 2 (IEEE, 2005), pp. 524–31.
25. The purpose here is not to explain how to use the LDA algorithm. This article simply aims to highlight problem space delineation as a key step in algorithm design in which human decision-making processes can enter into the structure of the algorithm. Many tutorials and explanations can be found online. The original article by Blei, Ng and Jordan is also accessible. See Blei, Ng and Jordan, 'Latent Dirichlet Allocation'.
26. Blei, Ng and Jordan, 'Latent Dirichlet Allocation', p. 995.

words are more important for a topic than others, one might also imagine a weighting scheme that gives more weight to more important words. In the topic 'pets', 'dog' and 'cat' would be accorded higher weightings than 'rain'. Though the word 'rain' might be included in a document about pets, it is less likely to be as common as 'dog' or 'cat'. *Topics are weighted combinations of words*. A document, then, *is a combination of topics*. It is a collection of vectors such as (00100), each weighted by their frequency.[27] The very large number of documents that need to be categorised by topic are transformed during the algorithm design process.

An algorithm designer must delineate a problem space and redefine a problem in order to allow the operation of an algorithm

The algorithm 'sees' each document as having been built from a combination of topics where each topic is a weighted combination of words. As it is not known which topics make up a document (they are latent), the LDA algorithm – which works backwards from the 'bag of words' view of each document to first find all of the combinations of words (topics) in that document and then determine the likelihood that the document was built from each combination of topics – can be used. Each document can then be assigned to the topics from which they were most likely to have been constructed. It is this final task – the calculation of the likelihood that a document falls within a topic category – that the algorithm really solves.[28] This is quite different, though analogous, to the way a human might think about and approach the task. The purpose of this transformation of the problem is to turn it into a task that a computer can solve within a strictly delineated problem space.

Every intelligence problem that is to be solved with the assistance of algorithms and AI must be transformed to some degree by the necessity of problem space delineation. The nature and extent of this transformation depends on the nature of the problem – highly formal problems might not be much transformed while qualitative tasks might be significantly transformed – and the decisions that are made during the design process bounded by the limits of human cognition, time, resources, legal frameworks[29] and data. An important practical implication of this is that intelligence agencies might know that a rival agency is using algorithms or AI to solve a particular problem. However, that problem is no longer exactly the same problem as it was before or as it would be if human decision-makers were tasked to solve it without the assistance of algorithms or AI.[30]

An algorithm designer must delineate a problem space and redefine a problem in order to allow the operation of an algorithm. A different delineation, however, would produce a different algorithm or solve a different problem. In finance, there are algorithms that treat investments as being solely represented by two statistics: average return; and standard deviation. The Markowitz critical line algorithm can be used to find portfolios that yield the highest average return at each level of standard deviation: the problem space is delineated by statistics amenable to optimisation and the investment problem becomes a matter of finding the optimal portfolio defined as the one with the highest return at some level of standard deviation.[31]

27. A document can be built from a single combination of words, that is, just one topic. For example, a document solely about 'medicine' is built from the topic called 'medicine', which is a combination of 'medical words'.
28. The LDA algorithm computes the most likely topics from which each document was constructed. It determines 'most likely' by applying the expectation-maximisation algorithm. This is the same 'engine' as that embedded within the PredPol predictive policing algorithm. See G O Mohler et al., 'Randomized Controlled Field Trials of Predictive Policing', *Journal of the American Statistical Association* (Vol. 110, No. 512, 2015), pp. 1399–411.
29. See Asaf Lubin, 'The Liberty to Spy', *Harvard International Law Journal* (Vol. 61, No. 1, 2020), pp. 185–244.
30. Even for a highly structured problem like playing a game of chess, some people have argued that chess algorithms do not play the same game as human chess players. Nathan Ensmenger, 'Is Chess the Drosophila of Artificial Intelligence? A Social History of an Algorithm', *Social Studies of Science* (Vol. 42, No. 1, 2011), pp. 5–30. This might explain why a chess player working with the assistance of a computer can defeat a chess algorithm. Garry Kasparov, *How Life Imitates Chess: Insights into Life as a Game of Strategy* (New York, NY: Bloomsbury, 2007). On this interesting aside, it is also possible to use an algorithm such as N Littlestone and M K Warmuth's weighted majority algorithm to guide one's choice, but this is subject to the same conditions that are outlined in this article. N Littlestone and M K Warmuth, 'The Weighted Majority Algorithm', *Information and Computation* (Vol. 108, No. 2, 1994), pp. 212–61.
31. Harry Markowitz, 'Portfolio Selection', *Journal of Finance* (Vol. 7, No. 1, 1952), pp. 77–91; Harry M Markowitz, *Portfolio Selection: Efficient Diversification of Investments* (New York, NY: John Wiley and Sons, 1959).

A designer who believes in company leadership above all else might deploy an algorithm to search for all positive news stories and accounts of every company's leaders with a view to investing only in those companies with 'quality leadership'. Such an investment algorithm would be completely different but equally valid, from a problem space delineation perspective. While breakthroughs in technology allow for problem spaces to be less strictly delineated, problem space delineation remains an inescapable constraint within which satisficing designers must operate. Because these constraints necessitate choices, designing effective algorithms and AI and countering those developed by rival agencies can be guided by decision theory and behavioural economics. These frameworks give us a list of things to think about if we want to understand the choices that algorithm designers have made (or have to make).

Heuristics and Algorithm Design

Beyond the foundational concepts of bounded rationality and satisficing, it has been established that people rely on heuristics when making decisions. Heuristics are rules of thumb that facilitate fast, intuitive problem solving. Heuristics are essential to decision-making because without them every problem would be subject to enormous 'deliberation cost'.[32] Despite this and despite certain advantages,[33] their use introduces systematic patterns into human choices, shaping estimates of outcomes and probabilities predictably over time. Although algorithms can be used to soften these patterns,[34] the use of heuristics might leave its mark on the very structure of algorithms. At the very least, decision theory provides some leads to follow up.

Tversky and Kahneman identified three heuristics that people tend to use.[35] These heuristics each replace a slow and deliberate assessment of probability with a quick assessment based on: (1) the ease with which instances of an event can be recalled (the availability heuristic); (2) how well a thing, person or event fits a particular category, such as the example of decision-makers assessing the likelihood that a particular person holds a particular profession by overlooking the prevalence or scarcity of jobs in the profession in favour of how well the person fits his or her idea of the characteristics of a worker in that profession (representativeness heuristic); and (3) a typically insufficient adjustment from an initial, potentially unconscious and unrelated numerical estimate (anchoring). Each of these factors replays systematically and continues to pull the decision-maker's probability estimates in predictable directions repeatedly over time. Much of this early research is summarised by Daniel Kahneman, Paul Slovic and Amos Tversky.[36]

> Decision theory and behavioural economics give us a list of things to think about if we want to understand the choices that algorithm designers have made (or have to make)

Developing their research programme, Kahneman and Tversky built a descriptive model of the decision-making process.[37] They called their model 'prospect theory'. In it, they incorporated several additional patterns of behaviour that had been observed. First, people tend to assess the outcomes of their options against a reference point. If a colleague has been given a two-step promotion, your one-step promotion might be viewed as a loss. Second, people are loss averse. They feel losses more than gains. Consequently, they take risks to

32. Conlisk, 'Why Bounded Rationality?', p. 671.
33. See Gerd Gigerenzer and Daniel G Goldstein, 'Reasoning the Fast and Frugal Way: Models of Bounded Rationality', *Psychological Review* (Vol. 103, No. 4, 1996), pp. 650–69.
34. Cass R Sunstein, 'Algorithms, Correcting Biases', *Social Research* (Vol. 86, No. 2, 2019), pp. 499–511.
35. This article was the culmination of their research up to that point. Daniel Kahneman and Amos Tversky, 'Subjective Probability: A Judgement of Representativeness', *Cognitive Psychology* (Vol. 3, No. 3, 1972), pp. 430–54; Amos Tversky and Daniel Kahneman, 'The Belief in the Law of Small Numbers', *Psychological Bulletin* (Vol. 76, No. 2, 1971), pp. 105–10; Amos Tversky and Daniel Kahneman, 'Availability: A Heuristic for Judging Frequency and Probability', *Cognitive Psychology* (Vol. 5, No. 2, 1973), pp. 207–32; Daniel Kahneman and Amos Tversky, 'On the Psychology of Prediction', *Psychological Review* (Vol. 80, No. 4, 1973), pp. 237–51.
36. Daniel Kahneman, Paul Slovic and Amos Tversky (eds), *Judgement Under Uncertainty: Heuristics and Biases* (New York, NY: Cambridge University Press, 1982).
37. Kahneman and Tversky, 'Prospect Theory'.

avoid losses. Third, people become risk averse once they have achieved gains. Finally, people tend to overweight unlikely outcomes and underweight likely outcomes. They do this even when they have the completely correct and objectively computed probabilities in hand.[38]

There are systematic ways in which the choices that must be made during algorithm design may be shaped by the use of heuristics and other characteristics of the human decision-making process. If they emerge, the patterns are the result of the designers' decisions in a dynamic, ever-changing context that is bounded by time, data availability, legal frameworks and human cognition. Thinking about decision-making in such a context and using decision theory and behavioural economics as a guide, at least several aspects of algorithm and AI design that might be shaped by the factors that this article has been considering can be identified. These not only relate to choices such as data and variables but to more fundamental choices such as the type of algorithm that will be built to solve a particular problem:

- Those features of the problem space that are most easily recalled may be accorded more prominence during design (availability). The availability heuristic can shape the choice of variables. For example, if a designer is developing a bankruptcy risk assessment algorithm, those variables that characterise individuals and companies most representative of bankruptcy may be accorded more prominence while other, potentially more relevant variables, may be excluded altogether. This is why some people are worried about discrimination arising from the use of algorithms.[39]
- Similarly, those features of the problem space that seem most representative of the core of the problem may be accorded more prominence. This may shape the choice of the type of algorithm to be built (representativeness).
- If parameters must be accorded values, the initial estimates or permissible ranges for these parameter values may be anchored (anchoring and adjustment).
- While algorithms do not feel losses more than gains – a true advantage of AI and algorithms over human decision-making – the designers of algorithms certainly do. Designing a bad algorithm results in career and prestige losses. Depending on the designer's organisational status and aspirations, the design may be far too speculative or far too conservative (loss aversion).
- Although the algorithm may compute or otherwise interact with probability estimates, the designer makes probability assessments before and during the design process. What are the chances that a particular algorithm structure can be made to work? What is the likelihood that it will be detected and countered? The answers to these questions may shape design decisions in ways that systematically reflect the weights that human decision-makers apply to the likelihood of outcomes (probability weighting).

While certain surface-level choices, such as which variables to include, may be shaped by the use of heuristics and other aspects of the human decision-making process, decision theory and behavioural economics leads to some of the more fundamental choices that designers make, including which type of algorithm to build to solve a particular problem. A deeper and more technical reliance on heuristics may emerge from the designer's experience and knowledge of particular algorithm structures. There is a relatively small algorithm design playbook that will form the starting point for many design attempts. The extent to which features of the problem to be solved are representative of a problem that has already been solved may combine with the ease with which the features of an existing algorithm structure can be recalled to shape the designer's choices from the very inception of the design project.

Prospect Theory and Counter AI

What are the practical implications of decision theory for designing better algorithms and countering a rival agency's algorithms and use of AI?

38. For a retrospective discussion of prospect theory, see Nicholas C Barberis, 'Thirty Years of Prospect Theory in Economics: A Review and Assessment', *Journal of Economic Perspectives* (Vol. 27, No. 1, 2013), pp. 173–96.
39. P Jeffrey Brantingham, Matthew Valasik and George O Mohler, 'Does Predictive Policing Lead to Biased Arrests? Results from a Randomised Control Trial', *Statistics and Public Policy* (Vol. 5, No. 1, 2018), pp. 1–6; Bruno Lepri et al., 'Fair, Transparent and Accountable Algorithmic Decision-Making Processes', *Philosophy and Technology* (Vol. 31, No. 4, 2018), pp. 611–27.

To think about this in a structured way, a more formal behavioural model of the decision-making process can be used. The most prominent of such models is Kahneman and Tversky's prospect theory,[40] which presents a framework that can be used to explore the broader decision-making context in which algorithms and AI are designed and built. This framework can also be used to think about the implications of the broader decision-making context for the specific technical tasks confronting designers. Intelligence agencies may be able to improve their own algorithm design by changing aspects of the organisational environment and they may improve the effectiveness of their efforts to counter rival algorithms and AI by identifying relevant aspects of the broader decision-making context characterising rival agencies.

Intelligence agencies may improve the effectiveness of their efforts to counter rival algorithms and AI by identifying relevant aspects of the broader decision-making context characterising rival agencies

Perhaps the central concept in Kahneman and Tversky's prospect theory is the reference point.[41] This is the idea that people do not always assess their alternatives in an absolute sense, but relative to a reference point. While Kahneman and Tversky treated the reference point as the status quo,[42] the reference point concept has since been expanded to include goals and aspirations. In making choices during the algorithm design process, the reference point may be the algorithmic structure that dominates practice within the designer's organisation, or has dominated the designer's education or past experiences. Markowitz's critical line algorithm that is widely used in finance was mentioned earlier. Like most economists, Markowitz already had a familiar mathematical structure – quadratic programming optimisation – in mind. If he could select just those features of the portfolio selection problem that could be made amenable to such a structure, he could use familiar optimisation techniques as the foundation for his algorithm. It is no surprise that an economist of Markowitz's generation would seek to build an algorithm around an optimisation technique.

It is also no surprise that seismological algorithms form the core of one of the most widely used predictive policing algorithms, PredPol.[43] George Mohler, one of the primary proponents of PredPol, completed a PhD at the University of California (Santa Barbara) where seismology is a prominent field of study. There are a relatively small number of categories of algorithms and, within those categories, popularly used designs. While the details can vary significantly, individual algorithms often rely on common specific characteristics in performing the problem solving task at hand.[44] One of the more popular categories of algorithms in the contemporary context is genetic algorithms (GA). These emerged in the 1960s, primarily due to the work of John Holland,[45] and his protégé, David Goldberg.[46] Goldberg later moved to the University of Illinois, which now houses the Illinois Genetic Algorithms Laboratory (IlliGAL). It would be completely unsurprising if an alumnus of this lab approached a new problem by delineating the problem space along the lines required by the GA structure. And a rival agency that recruits NLP algorithm and AI designers is not only embarking upon an information gathering and analysis task of some sort but is also signalling its reference point.

This is important because the reference point may influence decision-making in interesting

40. Kahneman and Tversky, 'Prospect Theory'.
41. *Ibid*.
42. *Ibid*.
43. G O Mohler et al., 'Randomized Controlled Field Trials of Predictive Policing', *Journal of the American Statistical Association* (Vol. 110, No. 512, 2015), pp. 1399–411.
44. See Steven S Skiena, *The Algorithm Design Manual* (London: Springer, 2008); Iztok Fister Jr et al., 'A Brief Review of Nature-Inspired Algorithms for Optimisation', *Elektrotehniški Vestnik* [*Electrotechnical Review*] (Vol. 80, No. 3, 2013), pp. 1–7; Arpan Kumar Kar, 'Bio Inspired Computing – A Review of Algorithms and Scope of Applications', *Expert Systems with Applications* (Vol. 59, 2016), pp. 20–32.
45. John H Holland, *Adaptation in Natural and Artificial Systems* (Cambridge, MA: MIT Press, 1992).
46. David E Goldberg, *Genetic Algorithms in Search, Optimization and Machine Learning* (Reading, MA: Addison-Wesley Professional, 1989).

ways. Psychologists and behavioural economists have found that people's preferences for risk vary depending on whether they find themselves in the domain of gains (above the reference point) or the domain of losses (below the reference point). When an algorithm design has been learned and applied with success, a design team that finds itself in the domain of gains may be risk averse and conservative, and less willing to make radical changes to its approach. A team that finds itself in the domain of losses for some reason, including a perception of lagging behind other agencies or simply because of the failure of previous application, may feel compelled to take more risk to recover lost ground. While the technical aspects of the design task may be directly shaped by the types of things that have been discussed in this article, the broader decision-making context is just as important. The designer's aspirations and past record of achievement may determine the position of the reference point and, therefore, the perspective from which current and future opportunities are evaluated.

If an intelligence agency believes that it is appropriate to view algorithm users/ designers as 'documents' and their choices of algorithms as 'words', the LDA algorithm could be trained to associate users/designers with particular algorithms

Intelligence agencies might use algorithms and AI to assist with keeping track of rival agencies' choices and, importantly, draw inferences about rival agencies' AI preferences. LDA, along with many other algorithms, can be used to infer preferences or, to be precise, can be used to solve a problem that is analogous to drawing inferences about people's preferences. Many of the algorithms that people use every day are capable of learning preferences – for example, Google's PageRank algorithm or Spotify's music recommendation algorithms.[47] Blei, Ng and Jordan explain how to apply LDA to infer people's preferences for movies. The problem space delineation is articulated with reference to the example discussed above: 'A user [viewer] and the movies chosen are analogous to a document and the words in the document (respectively)'.[48] If an intelligence agency believes that it is appropriate to view algorithm users/ designers as 'documents' and their choices of algorithms as 'words', the LDA algorithm could be trained to associate users/designers with particular algorithms. Then, when just one algorithm is known to be used by a particular user/designer (individual, team or agency), the LDA algorithm can be used to predict what other algorithms are likely to be in use. This would involve programming or 'training' the algorithm on a dataset of algorithms. It would also involve problem space delineation and a series of human choices.

Disinformation, Decision Theory and Counter-AI

Deception and disinformation have a long tradition in intelligence and military strategy, and disinformation was the core of Soviet active measures.[49] The use of disinformation has never gone away but its perceived importance oscillates over time. Alexander Babuta, Marion Oswald and Ardi Janjeva note the possibility that AI can be used to create volumes of synthetic disinformation.[50] This can reduce the overall quality of the information signals collected by a rival intelligence agency, increasing the likelihood of a 'disinformation cascade'.[51] AI can be used to carry out more effective cyber attacks

47. See Michelle Willson, 'Algorithms (and the) Everyday', *Information, Communication and Society* (Vol. 20, No. 1, 2017), pp. 137–50; Jakko Kemper and Daan Kolkman, 'Transparent to Whom? No Algorithmic Accountability Without a Critical Audience', *Information, Communication and Society* (Vol. 22, No. 14, 2019), pp. 2081–96.
48. Blei, Ng and Jordan, 'Latent Dirichlet Allocation', p. 1014.
49. Richard J Heuer Jr, 'Strategic Deception and Counter-Deception: A Cognitive Process Approach', *International Studies Quarterly* (Vol. 25, No. 2, 1981), pp. 294–327. See also Herbert Romerstein, 'Disinformation as a KGB Weapon in the Cold War', *Journal of Intelligence History* (Vol. 1, No. 1, 2001), pp. 54–67; Steve Abrams, 'Soviet Active Measures in Putin's Russia', *Connections* (Vol. 15, No. 1, 2016), pp. 5–31.
50. Alexander Babuta, Marion Oswald and Ardi Janjeva, 'Artificial Intelligence and UK National Security: Policy Considerations', *RUSI Occasional Paper* (27 April 2020).
51. Peter J Phillips and Gabriela Pohl, 'Disinformation Cascades, Espionage & Counter-Intelligence', *International Journal of Intelligence, Security and Public Affairs* (October 2020), DOI:10.1080/23800992.2020.1834311.

(that is, AI-based 'smart attacks') and to enhance defensive measures, especially against heavy data streams. This is a rapidly growing field of research.[52] An application of the LDA algorithm to infer rival agencies' preferences for unobserved algorithms could be countered by clouding the information about algorithm use, which is why traditional intelligence and detective work retain their place.

While disinformation in the context of algorithms and AI can be thought of as a purely technical matter, it is important to consider once more the effectiveness of (traditional) disinformation campaigns targeted against algorithm designers. The designers are human decision-makers and their choices influence the structure of the design and application of the algorithms and AI systems that they build. Disrupting the hostile use of algorithms and AI can be assisted by traditional intelligence regarding the background, status, aspirations and likely reference points of rival designers. Disturbing the reference point of rival algorithm designers in such a way that plunges them into the domain of losses or propels them into a falsely perceived domain of gains is a promising avenue for counter-AI. Defensively, decision theory helps to identify those parts of the human decision-making process that might be susceptible to being shaped by traditional or innovative disinformation.

Intelligence agencies are not the only ones dealing with a torrent of information that seemingly only becomes more turbulent over time. Algorithms and AI can be used to assist human decision-makers to sift through, dampen, exacerbate or hide within this turbulence. In approaching any of these tasks, decision theory and behavioural economics provide both a list of findings about human decision-making to consider as well as a framework to guide thinking about the implications of those findings in a particular context. This article has addressed algorithm and AI design. While technical considerations dominate much of the narrative, designers in rival agencies have made choices in developing their algorithms and AI that might be better understood (and better able to be countered) if relevant findings regarding human decision-making are considered. This article has pointed out just some of the implications of doing so. ■

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52. See Liang Xiao et al., 'IoT Security Techniques Based on Machine Learning: How Do IoT Devices Use AI to Enhance Security?', *IEEE Signal Processing Magazine* (Vol. 35, No. 5, 2018), pp. 41–49.